

SHAILJA Enterprises	TEST REPORT FOR TRANSFORMER FEEDER	SHEET : 1 OF 4
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BHILAI STEEL PLANT ,BHILAI (Chhattisgarh)		DATE : 29.04.2026

1. GENERAL

LOCATION- SUBSTATION: 26	CUBICAL NO. : 39	LCP NO. : 9137
CB NO: 45	FEEDER NAME: 26E – 2T 1000 KVA	

CIRCUIT BREAKER

2.1NAME PLATE DETAILS:-

Make	ASEA
Type	HCA 12/1240
Sl. No.	V0376
Rated Voltage	12 KV
Rated Current	1250 A
Control Voltage	110 VDC

2.2INSULATION RESISTANCE TEST: (MEASURED VALUES IN GΩ)

Reference	Breaker Condition	Applied Voltage (V)	R phase	Y phase	B phase
Pole – Pole'	Open	5000	>500	>500	>500
Phase – Phase	Close	5000	>500	>500	>500
Pole – Earth	Close	5000	>500	>500	>500

2.3COIL RESISTANCE TEST:

Terminal Reference	Measured Resistance in Ω
Closing coil	71.0
Tripping Coil	36.7
Spring charge motor	8.01 KΩ

2.4CONTACT RESISTANCE TEST:

Phase Ref.	Applied Current (100A)
	Measured Resistance in Micro ohms
R	78.0
Y	80.0
B	95.0

2.5TIME INTERVAL TEST:

Operation Ref.	Measured Time in milli Seconds		
	R	Y	B
Close	56	56	56
Open	40	40	40
Close – Open	48	48	48

CURRENT TRANSFORMER

3. RATIO TEST: (BY PRIMARY INJECTION)

3.1 CT (CTR: 100 / 5 A)

Phase Ref.	Injected current in Primary (A)	Measured current in Secondary (A) @ DISPLAY			
		RELAY	ENERGY METER	TRANSDUCER	AMMETER
R	100	5	-	-	100
Y	-	5	-	-	-
B	100	5	-	-	100

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SIGNATURE		

3.2 CBCT (RATIO: 150 / 1 A)

REFERENCE	Injected current in Primary (A)	Measured current in Secondary (mA)
CBCT	10	67.0

3.3 EARTH FAULT PROTECTION (E/F):- (THROUGH CBCT)

Phase	Injected Current (A)	Operated Time (Sec)
Io	1.8	OPERATED

RELAY

Make	ENGLISH ELECTRIC
Model(O/C,O/L,E/F)	CAG27AF8D/CAG39AF16A/CTU15AF8003AD(M)
Sl. No. (O/C,O/L,E/F)	M492863 / M492996 / M536636

4.0 PROTECTION TESTING: (SECONDARY INJECTION)

4.1 OVER CURRENT:

Set Current= 600 % Time=INST

Phase	Injected Current (A)	Status
R	32.6	OPERATED
B	29.9	OPERATED

4.2 OVERLOAD PROTECTION:

Setting= 5.0 A TDR= 0.8 Sec

Phase	Injected Current (A)	Operated Time (Sec)
R	5.3	0.899
Y	5.4	0.822
B	5.3	0.822

LT EARTH FAULT LT RELAY

MAKE	GENERAL ELECTRIC
MODEL	CDG11AF013SACH
SL. NO.	36265221 / 12 / 22

5.0 EARTH FAULT PROTECTION (E/F):

Setting= 4 A TMS= 0.15 sec

Phase	Rotation(A)	Operation(A)	Injected Current (A)			Operated Time (Sec)		
			X1.5	X2	X4	X1.5	X2	X4
RYB	4.5	4.6	6	8	16	0.926	0560	0.320

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TRANSFORMER TESTING

MAKE	BHEL
VECTOR GROUP	Dyn11
SL. NO.	2052971
RATING	1000 KVA
% IMPEDENCE	5.11 %
VOLTAGE RATIO	11000/440 V
CURRENT RATIO	52.49 /1312.16 A

6.1 WINDING RESISTANCE TEST:

Tap No	Measured value HV side in mΩ		
	1U-1V	1V-1W	1W-1U
NT	961.0	958.0	956.8

Measured value LV side Resistance in μΩ		
2u-2v	2v-2w	2w-2u
982.5	993.8	998.0
2u-2n	2v-2n	2w-2n
546.6	516.0	539.6

6.2 VOLTAGE RATIO TEST:

TAP NO.	HV side measured voltage in volts			LV side measured voltage in volts					
	1U1V	1V1W	1W1U	2u – 2v	2v – 2w	2w – 2u	2u – 2n	2v – 2n	2w – 2n
N	424.0	423.9	422.9	16.62	16.60	16.66	9.60	9.58	9.60

6.3 MAGNETIC BALANCE TEST: (At Normal Tap)

HV side measured voltage in volts		
1U-1V	1V-1W	1W-1U
425.1	317.0	130.0
238.0	423.0	185.7
67.9	356.9	422.0

6.4 MAGNETIZING CURRENT TEST:-

Applied Voltage in HV in volts				Measured Current in HV in Mille Amps		
Tap No	1U-1V	1V-1W	1U-1W	1U	1V	1W
NT	424.0	423.9	422.9	10.2	6.9	8.5

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LT BREAKER TESTING

7.1 NAME PLATE DETAILS:-

Make	NGEF
Type	ME2500
Sl. No.	4832510011
Rated Voltage	660 V
Rated Current	2500 A
Control Voltage	230 VAC

7.2 CONTACT RESISTANCE TEST:

Applied Current (100A)

Phase Ref.	Measured Resistance in Micro ohms
R	21.0
Y	25.0
B	23.0

8.0 FINAL CHECKS

S N	OBSERVATION	STATUS	RESULT
1	Complete Tightness Of Relay Panel	-	-
2	Relay Measurement Testing	DONE	OK
3	Relay Protection Testing	DONE	OK
4	Relay Indications (ON/OFF/Trip)	DONE	OK
5	Relay Communication (RXD & TXD)	-	-
6	Binary Input & Binary Output Checking	-	-
7	Cleaning of Auxiliary Relay	DONE	OK
8	Cleaning of PCB cards of Numerical Relay (if required)	DONE	OK
9	Consolidate data & Control Scheme Checked	DONE	OK
10	Trip of CB through Master Checked	DONE	OK
11	Field Simulation of OTI/WTI/BUCH Fault Checked./ Motor Checked	DONE	OK
12	All CT Links Closed & Tightness Check after Testing	-	-
13	All Protection Settings Normalized after Testing	DONE	OK
14	Testing DR Downloaded from the RELAY	-	-
15	Auxiliary Relay (HR & SR) operation check.	DONE	OK

9.0 REMARKS: 1) All test results are found satisfactory.

2) AR2 & IR2 relay operation check while shorting from marshalling box and TB 146-148 & TB 147-151 found ok from both sides.

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